

Antepartum

1. Big Picture of Pregnancy

Pregnancy is a **temporary, physiological, normal process** where all body systems adapt to support the growing fetus.

The nurse's job during antepartum care is not just to "know pregnancy facts." The nurse is constantly asking:

Is this normal pregnancy adaptation, or is this a danger sign?
Is mom okay? Is baby okay? What needs to be taught or reported?

Phase	Meaning
Antepartum / Prenatal	Before birth
Intrapartum	During birth
Postpartum	After birth

Pregnancy is divided into **three trimesters**, each about **13 weeks** long.

Main Nursing Goals During Pregnancy

Goal	What the Nurse is Focused on
Health teaching	Teaching the mother how to maintain good health
Risk identification	Finding risk factors early
Maternal monitoring	Monitoring vital signs, symptoms, weight, urine, labs
Fetal monitoring	Monitoring fetal growth, fetal heart rate, and fetal movement
High-risk support	Helping mothers with medical conditions improve health as much as possible
Safety teaching	Teaching warning signs that must be reported immediately

- **Smoking** increases risk for pregnancy complications and low birth weight.
- **Diabetes** increases risk for large babies, stillbirth, and placental/perfusion issues.

2. Confirming Pregnancy

Pregnancy signs are divided into three groups:

1. Presumptive
2. Probable
3. Positive

The key is knowing which signs actually confirm pregnancy.

Type	Meaning	Examples	Confirms Pregnancy?
Presumptive	Subjective symptoms reported by patient	Amenorrhea, nausea, breast tenderness, fatigue, urinary frequency, quickening	No
Probable	Objective signs, but could have another cause	Positive pregnancy test, Goodell's, Hegar's, Chadwick's, Braxton Hicks, ballottement, striae	No
Positive	Direct evidence of fetus	Fetal heartbeat, fetal movement felt by examiner, ultrasound visualization	Yes

Presumptive Signs of Pregnancy

These are **subjective**. The patient reports them.

Sign	Meaning
Amenorrhea	Missed period
Nausea	Common early pregnancy symptom
Breast tenderness	Hormonal breast changes
Deepening pigmentation	Darkening skin changes
Urinary frequency	Increased urination
Quickening	Mother feels fetal movement
Fatigue	Common early pregnancy symptom

NCLEX Tip: Presumptive signs are not enough to confirm pregnancy because stress, illness, weight changes, and hormonal issues can also cause them.

Probable Signs of Pregnancy

These are **objective**, but still do not fully confirm pregnancy.

Sign	Meaning
Goodell's sign	Softening of cervix
Hegar's sign	Softening of lower uterine segment
Braxton Hicks contractions	Irregular contractions
Ballottement	Examiner pushes fetus and fetus rebounds
Striae	Stretch marks
Positive pregnancy test	hCG present
Chadwick's sign	Bluish/purple discoloration from increased vascularity

Important: A positive pregnancy test is **probable**, not positive. It shows hCG, but it does not directly show the fetus.

Positive Signs of Pregnancy

These are the only signs that confirm pregnancy.

Positive Sign	Why It Confirms Pregnancy
Audible fetal heartbeat	Fetal life is heard
Fetal movement felt by examiner	Examiner feels fetal movement
Ultrasound visualization	Fetus is seen

Normal fetal heart rate: 110–160 bpm

3. Pregnancy Terms and GTPAL

Term	Meaning	Key Point
Gravida	Total number of pregnancies	Includes current pregnancy
Para	Births at 20 weeks or greater	Does not depend on whether baby is living now
Age of viability	Point where fetus may survive outside uterus	Usually around 20-24 weeks, but varies

GTPAL System

GTPAL describes pregnancy history.

Letter	Meaning
G	Gravida: total pregnancies
T	Term births, greater than 37 weeks
P	Preterm births, less than 37 weeks
A	Abortions, spontaneous or induced
L	Living children

Important Multiples Rule

Twins count as **one pregnancy** for gravida.

Example: A woman pregnant once with twins is still **G1**, not **G2**.

4. Naegele's Rule

Naegele's Rule is used to estimate the due date.

Step	What To Do
1	Start with the first day of the last menstrual period
2	Count back 3 months
3	Add 7 days
4	Correct the year if needed

Example

Step	Date
LMP	April 2, 2026
Count back 3 months	January 2, 2026
Add 7 days	January 9, 2026
Correct year	January 9, 2027

5. Initial Prenatal Visit

The first prenatal visit establishes **baseline data**. This is where the nurse/provider figures out where the patient is starting from before monitoring changes throughout pregnancy.

Initial Prenatal Visit: What Happens

Category	What Is Assessed
Physical assessment	Complete physical exam
Reproductive assessment	Pelvic exam; evaluation of pelvis and reproductive organs
Pregnancy dating	Estimated date of delivery, ultrasound if needed
Labs	Routine prenatal labs
Risk review	Medical, pregnancy, social, nutritional, and safety risk factors

Lab / Screening	Why It Matters
Hemoglobin / hematocrit	Screens for anemia
wBC count	Infection clue, though wBC can be elevated in pregnancy
Platelets	Bleeding/clotting concerns
Blood type	Needed for transfusion/Rh status
Rh factor	Determines need for RhogAM
Rubella status	Determines immunity
Gonorrhea/chlamydia	Infection risk to mom/baby
Syphilis	Can harm fetus
HIV	Treatment reduces transmission risk
Hepatitis B	Newborn protection planning
Urinalysis	Protein, glucose, ketones, infection
Group B strep status if known	Labor antibiotic planning

Rh Factor and RhogAM

If the mother is **Rh negative**, she may need **RhogAM** depending on the situation.

Situation	Why
Around 28 weeks	Prevent sensitization during pregnancy
After miscarriage	Possible fetal blood exposure
After trauma	Possible fetal-maternal blood mixing
After invasive procedures	Possible fetal blood exposure
Postpartum if baby is Rh positive	Prevent problems in future pregnancies

Prenatal Appointment Schedule

Gestational Age	visit Frequency
Conception to 28 weeks	Every 4 weeks
29-36 weeks	Every 2-3 weeks
37 weeks to birth	weekly

6. Routine Assessment at Each Prenatal visit

At every prenatal visit, think:

BP, weight, pee, baby.

That simple phrase keeps the visit organized.

Routine visit Assessment Chart

Category	what to Check	why it Matters
BP	Same arm, same position each time	BP should not normally rise; rising BP may suggest preeclampsia
weight	Pattern of weight gain	Too much or too little can signal problems
urine	Protein, glucose, ketones	Screens for preeclampsia, gestational diabetes, poor intake/dehydration
Baby	Fundal height, FHR, movement, Leopold's maneuvers	Checks fetal growth, position, and well-being
Risk factors	Old and new risks	Pregnancy risk can change over time
Nutrition	Diet, intake, supplements	Supports maternal/fetal health
Discomforts	New or worsening symptoms	Determines normal discomfort vs danger sign

Urine Findings

Finding	Possible Concern
Protein	Possible preeclampsia
Glucose	Possible gestational diabetes
Ketones	Poor intake, dehydration, starvation state

Fundal Height

Fundal height helps determine:

- whether fetus is growing as expected
- Possible amniotic fluid issues
- Possible abnormal fetal size

In general, after about **20 weeks**, fundal height in centimeters should roughly equal weeks of gestation.

Example: 24 weeks pregnant $\approx$ 24 cm fundal height

7. Normal Physiological Changes in Pregnancy

What is normal, and what would make it unsafe?

System	Normal Pregnancy Changes	Nursing Concern
Reproductive	uterus grows, cervix softens, mucus plug forms	Bleeding, infection, preterm cervical change
Vagina	Leukorrhea, acidic pH, increased vascularity	Foul odor, itching, burning, abnormal discharge
Breasts	Tenderness, fullness, colostrum	Avoid nipple stimulation unless instructed
Respiratory	Mild dyspnea, increased O ₂ needs, congestion	Severe SOB, chest pain, cyanosis
Cardiovascular	Increased blood volume, cardiac output, clotting factors	HTN, clots, supine hypotension, heart failure worsening
GI	Nausea, heartburn, constipation, slowed motility	Persistent vomiting, dehydration
Urinary	Frequency, ureter dilation, increased UTI risk	UTI/pyelo can trigger preterm labor
Fluid/Electrolytes	Sodium retention, edema	Face/hand swelling, water intoxication
Skin	Striae, linea nigra, melasma, spider nevi	Usually cosmetic/normal

Reproductive System

Uterus

During pregnancy, the uterus:

- Increases in length
- Increases in depth
- Increases in width
- Increases in weight
- Wall thickens
- Volume increases
- Blood flow increases

The uterus size is similar to an **upside-down pear** prior to pregnancy.

Cervix

The cervix becomes:

- More vascular
- Edematous
- Softer

Term	Meaning
Operculum	Mucous plug that helps seal out bacteria
Effacement	Thinning of cervix, measured 0-100%
Dilation	Opening of cervix, measured 0-10 cm
Goodell's sign	Softening of cervix
Chadwick's sign	Bluish/purple discoloration of cervix and vagina

Mucous Plug

The mucous plug protects against bacteria during pregnancy.

Effacement and Dilation

Effacement usually occurs before major dilation, especially in a first-time pregnancy.

- **Effacement** = cervix thins
- **Dilation** = cervix opens from 0-10 cm

vagina

Normal changes:

- Leukorrhea
- More acidic pH
- Increased vascularity
- Chadwick's sign

Leukorrhea

Leukorrhea is a white vaginal discharge that can occur throughout pregnancy.

Teaching

- This can be normal.
- Report foul odor, itching, burning pain, or green/yellow discharge.

Acidic pH

The vagina becomes more acidic, which can help reduce some infections but may increase yeast infections.

Breasts

Normal changes:

- Fullness
- Tingling
- Tenderness
- Areola darkening/enlargement
- Colostrum may appear by 16 weeks

Colostrum

Feature	Description
Appearance	Thin, watery
Nutrition	High in protein
Immune benefit	Rich in maternal antibodies
Role	Precursor to breast milk
Timing	Secreted during first 2-3 days after birth
Teaching	Encourage mother to breastfeed at least once so baby benefits

Important: Do not express/pump colostrum during pregnancy unless specifically instructed, because breast stimulation can trigger uterine contractions.

Respiratory System

Normal Change	Why It Happens / Teaching
Mild shortness of breath	Can happen as oxygen demand increases
Oxygen consumption increases 15-20%	Pregnancy increases maternal/fetal oxygen needs
Nasal stuffiness/congestion	Estrogen can cause congestion

NCLEX Tip: Mild dyspnea can be normal, but severe shortness of breath, chest pain, cyanosis, or respiratory distress is not normal.

Cardiovascular System

Pregnancy causes a major increase in circulating volume. The body is working harder to perfuse both mom and baby.

Change	Meaning / Why It Matters
Blood volume increases by 50% or 1.5x normal	Supports placenta and fetus
Cardiac output increases	Heart works harder
Iron needs increase	Needed for RBC production
Physiological anemia can occur	Plasma volume increases more than RBCs, diluting blood
Palpitations may occur	Heart workload increases
Edema and varicosities may occur	Venous return is slowed
Clotting factors increase	Body prepares for birth, but clot risk increases
Functional murmur may be heard	Increased cardiac workload can make murmur noticeable. Might require Echocardiogram.
BP normally does not rise	Rising BP is concerning
Pregnancy worsens heart failure	Increased volume increases cardiac workload
Blood lipids falsely elevated	Do not draw routine lipid levels during pregnancy
Orthostatic hypotension common	Vessels dilate to accommodate increased volume

Iron and Physiological Anemia

Physiological anemia occurs because plasma volume increases more than red blood cells. The blood becomes diluted, so hemoglobin/hematocrit may look lower. Monitor and give body time to catch up.

Iron is in prenatal vitamins, but the patient may require an additional supplement if she has a history of iron deficiency anemia.

Common complaints with iron:

- Nausea
- Constipation

Iron Deficiency Anemia

Anemia if hemoglobin is less than **11 g/dL**.

Signs/symptoms may include:

- Fatigue
- Weakness
- Dizziness
- Pallor
- Shortness of breath

Edema, varicosities, and Clot Risk

Issue	Teaching / Nursing Concern
Edema	Elevate legs above heart
Varicose veins	Increase risk of blood clots
TED hose	May be used prophylactically especially with hx. varicose veins.
Hemorrhoids	Can occur from varicosities/pressure
Face/finger swelling	Concerning for preeclampsia
Calf pain, warmth, redness	Report; possible clot
Sudden SOB/chest pain	Emergency; possible pulmonary embolism

Redheads more at risk for bleeding.

Blood Pressure

Blood pressure normally does **not** rise during pregnancy.

NCLEX Tip: A rising blood pressure in pregnancy is concerning for hypertensive disorders, especially preeclampsia.

Supine vs Orthostatic Hypotension

Type	Cause	Symptoms	Nursing Teaching
Supine hypotension	Pregnant uterus compresses inferior vena cava	Lightheadedness, faintness, palpitations, dizziness	Avoid lying flat; side-lying, preferably left
Orthostatic hypotension	Dilated vessels + position change	Dizziness when standing	Change positions slowly; dangle feet before standing

Why Supine Hypotension Matters

Compression of the vena cava decreases venous return. This can decrease cardiac output and reduce blood flow to the fetus.

Gastrointestinal System

Pregnancy hormones slow down the GI tract. Food moves more slowly, and the growing uterus pushes on the stomach and intestines. This is why GI discomforts are so common.

Change	Why It Happens / Concern
Nausea and vomiting	Hormonal changes and slowed GI movement
Heartburn	Slowed emptying + pressure from uterus
Constipation	Slowed peristalsis
Flatulence	Slowed GI motility
Gingivitis/bleeding gums	Increased estrogen/hyperemia
Gallstones	Bile changes and decreased emptying
Itching/jaundice	Possible bile-related changes

Urinary System

Normal urinary changes:

- Glycosuria may be more common.
- Proteinuria may be more common.
- Ureters increase in diameter.
- Ureters lose tone.
- Frequency increases in first and last trimesters.
- Risk for bladder infection and pyelonephritis increases.

The bladder holds more, so the patient should empty the bladder frequently to prevent back flow into the kidneys.

Why UTIs Matter in Pregnancy

UTIs can trigger **preterm labor**.

Report This	Why
Burning with urination	Possible UTI
Frequency with pain	Possible UTI
Fever	Infection concern
Back/flank pain	Possible pyelonephritis
Cloudy or foul-smelling urine	Infection concern

Urine Collection

- Clean catch urine **every time (avoid contamination/false positives)**
 - wash hands before collecting the specimen.
 - open the sterile cup without touching the inside of the cup or lid.
 - Separate the labia with one hand.
 - Clean the urinary area from **front to back** using the provided wipes.
 - Begin urinating into the toilet first.
 - After the urine stream has started, place the cup into the stream.
 - Collect the needed amount of urine.
 - Remove the cup before finishing urination.
 - Finish urinating into the toilet.

- Secure the lid tightly without touching the inside of the cup.
- Give the specimen to the nurse promptly.
- If patient cannot obtain clean catch specimen, an in/out catheter may be needed.

Fluid and Electrolyte Balance

Normal changes:

- Sodium retention can cause edema.
- Edema can contribute to carpal tunnel syndrome.

Pitocin and water Intoxication

Pitocin has an antidiuretic effect, meaning it can cause the body to retain water.

Possible Sign	Concern
Agitation	water intoxication
Delirium	water intoxication
Confusion	water intoxication
Headache	water intoxication
Decreased urine output	Fluid retention
Seizures	Severe water intoxication

W/O is monitored during labor and the immediate postpartum period.

Integumentary System

Change	Meaning
Striae gravidarum	Stretch marks, usually genetic/from rapid weight gain
Diastasis recti	Separation of rectus abdominal muscles
Linea nigra	Dark line from umbilicus to pubis
Melasma/chloasma	Mask of pregnancy; darker skin around eyes; common in certain ethnicities
vascular spiders/spider nevi	Small visible blood vessels, usually on thighs
Palmar erythema	Redness/itching of palms

Most pigment changes are caused by increased estrogen and fade after pregnancy.

8. Common Discomforts of Pregnancy and Teaching

Discomfort	Teaching
Nausea	Dry toast/crackers before getting out of bed; fluids between meals; small frequent meals; avoid greasy, spicy, strong odor foods
vaginal discharge	Daily bathing; loose cotton underwear; wipe front to back; no douching unless ordered
Fatigue	Sleep 8-10 hours/night; nap if possible; promote rest
Backache	Maintain posture; low-heeled shoes; squat instead of bending; use pillows for support
Constipation	8 glasses water/day; fiber; exercise; bowel schedule; limit cheese/sweets; laxatives/enemas only if ordered
Varicose veins	Avoid tight clothing/crossing knees; elevate legs; support hose; avoid standing in one position too long
Hemorrhoids	Sitz bath; witch hazel; anesthetic ointment if approved; avoid constipation
Heartburn	Small frequent meals; avoid fatty foods; reduce caffeine; avoid smoking; sit upright after meals; sleep elevated; antacids if approved
Nasal stuffiness	Saline drops; cool mist humidifier only
Dyspnea	Rest with upper torso propped up; avoid overexertion
Leg cramps	Extend affected leg; keep knee straight; dorsiflex foot; elevate legs periodically
Lower extremity edema	Rest; elevate legs above heart; avoid tight clothing; monitor for sudden swelling of face/hands

Normal vs Concerning Edema

Normal	Concerning
Mild ankle/leg edema	Swelling of face
worse after standing	Swelling around eyes
Improves with rest/elevation	Sudden swelling of hands
No other symptoms	Headache, vision changes, RUQ pain

Mild ankle/leg edema can be normal. Swelling of the **face, hands, or around the eyes** is more concerning and may indicate preeclampsia.

9. Danger Signs of Pregnancy

These findings should be reported immediately because they can point to hemorrhage, infection, preeclampsia, fetal distress, dehydration, or preterm labor.

Danger Sign	Possible Concern
Vaginal bleeding with or without pain	Placenta issue, miscarriage, hemorrhage
Rupture of membranes/leaking fluid	PROM, infection risk, cord prolapse risk
Swelling of fingers, face, or around eyes	Preeclampsia
Visual disturbances	Preeclampsia
Persistent/severe abdominal pain	Labor, abruption, emergency
Chills or fever	Infection
Painful urination	UTI/pyelonephritis
Persistent vomiting	Dehydration, electrolyte imbalance
Change in fetal movement	Possible fetal distress

Decreased Fetal Movement

If baby is not moving:

1. Have mother drink something cold.
2. Have mother eat if she has not eaten.
3. Have mother lie on side.
4. Have mother focus on fetal movement.
5. If movement does not improve, call provider/go in for evaluation.

10. Psychological Adaptation to Pregnancy

Psychological adaptation works best chronologically by trimester.

Trimester	Main Emotional Task	Common Findings
First trimester	Accepting the pregnancy	Ambivalence
Second trimester	Accepting the baby	Baby feels more real; quickening; follows prenatal instructions
Third trimester	Preparing for parenthood	Nesting, prenatal classes, preparing for birth, making plans for baby

Ambivalence

Ambivalence means feeling both pleased and not pleased about the pregnancy. This can be normal.

Prenatal Classes

Prenatal classes are most beneficial during the **third trimester** because the information is more fresh.

11. Nutrition During Pregnancy

Pregnancy increases nutritional needs, but the patient does not need to "eat for two." The increase is more specific.

Situation	Extra Calories needed
Pregnancy	300 extra calories/day
Breastfeeding	500 extra calories/day

Breastfeeding mothers need about **200 calories and 5 g protein more** than pregnant women.

Weight Gain

Time	Expected weight gain
First trimester	2-3 lb total
After first trimester	About 1 lb per week

Recommended Total weight gain

Prepregnancy weight	Recommended Total gain
Normal BMI	25-35 lb
Underweight	28-40 lb
Overweight	15-25 lb
Obese	About 15 lb
Twin pregnancy	35-45 lb

Iron

Iron supplements are often prescribed.

Common dose: 30 mg/day

Teaching Point	Why It Matters
Best absorbed on empty stomach	Improves absorption
Take with orange juice	Vitamin C helps absorption
Do not take with coffee, tea, or milk	Decreases absorption
May cause nausea/heartburn/epigastric pain	Common side effects
May cause constipation or diarrhea	Common side effects
May cause black stools	Expected finding
Keep away from children	Iron overdose can be fatal
Liquid iron: use straw and rinse mouth	Prevents staining teeth

Folic Acid

Folic acid helps prevent neural tube defects.

Low folic acid increases risk for:

- Spina bifida
- other neural tube defects

Alcohol use can increase risk for folic acid deficiency.

Nutrition Risk Factors

Risk Factor	Why It Matters
Culture	Affects food choices
Age	Adolescents may have different nutritional needs
Nutritional knowledge	Affects choices and supplement use
Socioeconomic status	Healthier foods may be more expensive
Adolescence	May eat less to conceal pregnancy
Vegetarianism	May need protein/iron/B12 planning
Lactose intolerance	More common in African Americans
Nausea/vomiting	Can reduce intake
Anemia	May be associated with pica
Abnormal prepregnancy weight	Affects recommended gain
Eating disorders	Can affect maternal/fetal nutrition
Multiparity	May increase nutritional demands
Substance use/abuse	Can impair nutrition and fetal growth

Pica

Pica means craving or eating non-food substances.

Examples:

- Dirt
- Clay
- Ice
- Starch

Pica may be associated with iron deficiency. Report it because it may indicate nutritional deficiency and can expose mother/fetus to harmful substances.

12. Prenatal Diagnostic Testing

Prenatal diagnostic tests are used to:

- Detect congenital anomalies
- Monitor fetal well-being in high-risk pregnancy
- Provide accurate gestational age
- Guide interventions

Indications for Prenatal Testing

Indication	Why It May Matter
Diabetes	Placental/perfusion and fetal growth concerns
Maternal age greater than 35	Higher chromosomal risk
Previous infant over 4000 g	May suggest diabetes/fetal growth concerns
Excessive weight gain	May suggest complications
Previous chromosomal abnormality	Increased genetic risk
Family history of chromosomal abnormality	Increased genetic risk
Previous neural tube defect	Increased recurrence risk
Three or more spontaneous abortions	Possible genetic/maternal issue
Elevated maternal serum AFP	Possible fetal anomaly
Maternal Rh sensitization	Fetal hemolytic disease concern

ultrasound

ultrasound visualizes the fetus.

used For	why
Gestational age	Dating pregnancy
Fetal growth	Checks size
Fetal position	Presentation
Placenta location	Rules out location concerns
Amniotic fluid	Too much/too little fluid
Fetal anomalies	Anatomy assessment
Multiple gestation	Twins/multiples

Full Bladder Teaching

Before the second trimester, the uterus is still low in the pelvis. A full bladder may be needed to act as a "window" for visualization.

Later in pregnancy, this may not be needed because the uterus rises into the abdomen.

AFP Screening

AFP stands for **alpha-fetoprotein**.

It can be measured in:

- Maternal serum: **MSAFP**
- Amniotic fluid: **AFAFP**

AFP Result	Possible Meaning
High AFP	May indicate open neural tube defect
Low MSAFP	May be associated with chromosomal anomalies such as trisomy 21

Important: AFP is a **screening test**, not diagnostic. Abnormal AFP requires follow-up testing, such as high-resolution ultrasound or amniocentesis.

Multiple Marker Screening

Multiple marker screening may include:

Marker

AFP

hCG

Unconjugated estriol

Inhibin A

It helps increase detection of:

- Trisomy 18
- Trisomy 21

Important: This is also **not diagnostic**. It identifies risk and need for further testing.

Amniocentesis

Amniocentesis is aspiration of amniotic fluid. It is ultrasound-guided and performed by a physician.

Category Notes

Sample Amniotic fluid

Timing First/second trimester for genetic testing; third trimester for lung maturity/hemolytic disease

Results May take 2-4 weeks

Main risk Spontaneous contractions, infection, bleeding, leaking fluid, pregnancy loss

Uses

Amniocentesis can be used to:

- Examine fetal cells
- Identify genetic abnormalities
- Detect high AFP
- Evaluate fetal condition if mother is Rh sensitized
- Diagnose amnionitis/intrauterine infection

- Test amniotic fluid AFP after abnormal MSAFP
- Determine fetal lung maturity in third trimester
- Determine fetal bilirubin if Rh incompatibility is present

Nursing Considerations

Before Procedure	After Procedure
Explain procedure	Monitor fetal heart rate
verify informed consent	Monitor for contractions
Assess fetal heart rate	Monitor for vaginal bleeding
Assist with positioning	Monitor for leaking fluid
Ensure bladder status is appropriate	Teach to report fever, chills, bleeding, leaking fluid, or contractions

Chorionic Villus Sampling

CVS = diagnostic prenatal test that samples chorionic villi from the placenta.

It may not be covered by insurance.

Category	Notes
Sample	Chorionic villi from placenta
Timing	10-13 weeks
Advantage	Earlier than amniocentesis
Results	Diagnosis may be available in 2-4 days
Important use	Helps with earlier decision-making/possible earlier termination
Rh-negative patient	May need RhogAM after CVS
Risk	Can be associated with limb defects if performed too early

used to Diagnose

- Down syndrome
- Trisomy 18/13
- Sickle cell disease

- Cystic fibrosis
- Tay-Sachs

Nursing Priorities

Before	After
verify informed consent	Monitor for bleeding
Check maternal VS	Monitor for cramping
Assess fetal status if appropriate	Monitor for leaking fluid
Check Rh status	Monitor for fever/chills and infection signs

Fetal Lung Maturity

Amniocentesis can check fetal lung maturity if delivery is being considered before **38 weeks**.

L/S Ratio

L/S ratio refers to:

- Lecithin
- Sphingomyelin

A ratio of **2:1** generally suggests fetal lung maturity.

Why It Matters

Surfactant keeps alveoli open by reducing surface tension.

without enough surfactant:

- Alveoli collapse
 - Respiratory distress can occur after birth
 - Pneumothorax, or lung collapse
-

13. Antepartum Fetal Surveillance

These tests evaluate fetal well-being before birth.

Test	what it Evaluates	Good/Reassuring Finding
Nonstress test	FHR response to fetal movement	Reactive accelerations
Contraction stress test	FHR response to contractions	No concerning decelerations
Biophysical profile	Five fetal well-being parameters	8-10
Modified BPP	AFI + NST	Normal fluid and reactive NST
Kick counts	Maternal perception of fetal movement	Adequate movement

Nonstress Test

NST evaluates fetal heart rate response to fetal movement.

Purpose

To check if the fetal heart rate accelerates with movement.

This suggests:

- Adequate oxygenation
- Healthy neural pathway

NST Results

Result	Meaning
Reactive	Reassuring
Nonreactive	Expected accelerations are not present; baby may be sleeping or further testing may be needed

Reactive NST Criteria

A reactive NST means:

- At least 2 accelerations
- Each acceleration is at least 15 bpm above baseline
- Each lasts at least 15 seconds
- Occurs within a 20-minute period

Nonreactive NST: Possible Next Steps

- Continue monitoring
- Stimulate fetus
- Give mother food/fluids if appropriate
- Further testing such as CST or BPP

Easy Memory: NST = looking for accelerations. Accelerations are good.

Placement Note

For fetal monitoring, one monitor is placed to trace **fetal heart rate**, and the other is placed to trace **uterine activity/contractions**. The exact placement depends on fetal position and where the strongest fetal heart tones are heard.

Contraction Stress Test

CST evaluates how the fetus responds to contractions.

Purpose

During contractions, fetal oxygenation can decrease. The CST checks whether the fetus can tolerate that temporary stress.

When used

May be done if NST is nonreactive.

How Contractions Are Caused

- Nipple stimulation
- Low-dose oxytocin

CST Results

Result	Meaning
Negative CST	Reassuring; no concerning late decelerations
Positive CST	Bad; decelerations occur with 50% or more contractions

Important NCLEX Point: Positive CST is bad. It means the fetus may not tolerate labor well.

Oxytocin Safety

- Always piggyback oxytocin into the main IV line.
- Never run as the main bag because too much could cause excessive contractions.

Monitor for:

- Uterine tachysystole
- Fetal distress
- Late decelerations

NST vs CST

Test	what it Measures	Good Result	Bad Result
NST	FHR response to fetal movement	Reactive accelerations	Nonreactive
CST	FHR response to contractions	No concerning decels	Decels with $\geq 50\%$ contractions

Memory Trick:

NST: accelerations are good

CST: positive is bad

Biophysical Profile

BPP evaluates fetal well-being during the antepartum period.

It assesses five parameters:

1. Nonstress test
2. Fetal breathing movements
3. Gross fetal movements
4. Fetal tone
5. Amniotic fluid volume

BPP Scoring

Each category can receive:

- **2 points** = normal
- **0 points** = abnormal

Maximum score: **10**

Score	Meaning
8-10	Reassuring
6	Equivocal
4 or less	Abnormal

A normal BPP can allow conservative treatment of high-risk patients and may delay delivery if fetal well-being is reassuring.

BPP Parameters

Parameter	Normal
NST	Reactive
Fetal breathing	At least 1 episode in 30 min
Gross body movement	At least 3 movements in 30 min
Fetal tone	At least 1 episode of extension/flexion
Amniotic fluid	At least one adequate pocket of fluid

Modified Biophysical Profile

Modified BPP may use only:

- Amniotic fluid index
- Nonstress test

It is a shorter way to assess fetal well-being.

Maternal Assessment of Fetal Movement

Also called **kick counts**.

This is done by the mother. Fetal movement is associated with fetal condition.

Kick Count Notes

Topic	Notes
who does it?	Mother
How often?	May be done 30 minutes, 3 times per day
when to evaluate further	Fewer than 4 movements in 30 minutes to 1 hour

Some mothers may not feel movement as strongly, and babies also sleep. However, decreased movement should still be evaluated and should not be ignored.

Simple Prenatal Testing

Situation	Tests
Routine pregnancy / first visit	CBC, blood type/Rh, antibody screen, rubella titer, HIV, hepatitis B, syphilis, gonorrhea/chlamydia, urinalysis
Routine pregnancy / every visit	BP, weight, urine dip, fundal height, fetal heart rate
Routine pregnancy / 16-18 weeks	AFP or multiple marker screening may be offered
Routine pregnancy / 24-28 weeks	Glucose screening
Rh-negative mother	RhoGAM
Uncertain due date or fetal growth concern	Ultrasound
Genetic/chromosomal risk	CVS or amniocentesis
Abnormal AFP or multiple marker screening	Ultrasound and/or amniocentesis
Need fetal lung maturity before early delivery	Amniocentesis
High-risk pregnancy or decreased fetal movement	NST or BPP
Nonreactive NST	BPP or CST
Concern baby will not tolerate contractions	CST

14. Diagnostic Testing Considerations

Parents are often concerned about:

- Safety of the fetus
- Duration of procedure
- How long results take
- Cost
- Insurance coverage
- Discomfort to mother

Nursing Role

The nurse should:

- Provide clear teaching
- Reinforce provider explanation
- verify informed consent when needed
- Assess emotional response
- Avoid giving false reassurance
- Encourage questions

15. TORCH Infections

TORCH infections can cross the placenta and harm the fetus. These infections can be especially harmful in the first trimester because fetal organs are developing.

letter	Infection
T	Toxoplasmosis
O	other infections
R	Rubella
C	Cytomegalovirus
H	Herpes simplex virus

Toxoplasmosis

often associated with exposure to:

- Cat feces/litter box
- undercooked meat
- Contaminated soil

Teaching

- Avoid changing litter box if possible.
- wear gloves when gardening.
- wash hands well.
- Cook meat thoroughly.

Herpes Simplex Virus

If active genital lesions are present at birth, cesarean birth may be needed.

Reason:

- To prevent neonatal exposure during vaginal birth

Rubella

Rubella may cause mild illness in the mother but serious fetal effects.

Rubella overview

Category	Findings
Maternal effects	Mild rash, mild systemic illness
Fetal effects	Deafness, mental challenges, motor challenges, cataracts, cardiac defects, PDA, pulmonary stenosis, IUGR/SGA, thrombocytopenic purpura, cleft lip/palate
Immunity	Titer greater than 1:8 suggests immunity
Vaccine	Not given during pregnancy
Postpartum	May be given postpartum if mother is not immune
Newborn	Infants born to mothers who had rubella may need isolation in nursery

Quick Final Review Table

Topic	Must Remember
Positive pregnancy test	Probable sign, not positive sign
Positive signs	FHR, movement felt by examiner, ultrasound
Normal FHR	110-160 bpm
BP in pregnancy	Should not normally rise
Supine position	Avoid due to vena cava compression
Protein in urine	Possible preeclampsia
Glucose in urine	Possible gestational diabetes
Ketones in urine	Poor intake/dehydration
NST	Reactive accelerations are good
CST	Positive is bad
BPP	8-10 reassuring, 6 equivocal, ≤ 4 abnormal
Rubella vaccine	Do not give during pregnancy
Iron	Take with vitamin C; avoid milk/coffee/tea
Decreased fetal movement	Always evaluate